

TOPEKA BILLARD, KS, USA

WMO: 724560

Lat: 39.0722N

Lon: 95.6306W

Elev: 881

StdP: 14.23

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13996

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.0	8.9	-5.5	4.2	6.7	-1.4	5.3	10.8	25.3	34.0	23.0	34.1	7.5	310	0.507

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.9	98.2	76.2	94.9	76.1	92.0	75.2	79.3	91.8	78.0	90.4	76.9	88.8	10.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.8	139.3	86.2	74.5	133.2	84.9	73.3	127.9	83.8	43.4	91.9	42.1	90.6	40.9	88.7	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.8	19.8	17.9	-3.5	102.3	4.8	3.4	-7.0	104.8	-9.8	106.8	-12.6	108.7	-16.1	111.1
			-4.3	81.7	4.7	1.4	-7.7	82.7	-10.4	83.6	-13.0	84.4	-16.4	85.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	56.1	30.3	34.8	45.5	55.7	65.9	75.6	79.9	78.1	69.6	57.4	44.8	33.8
	DBStd	19.56	11.32	11.74	11.47	9.43	8.21	6.24	5.97	6.11	8.07	9.32	9.80	10.43
	HDD50	2081	613	441	221	48	2	0	0	0	1	35	210	510
	HDD65	4823	1075	847	611	302	93	4	0	1	43	271	608	968
	CDD50	4294	3	14	81	220	495	768	926	872	590	264	54	7
	CDD65	1557	0	0	5	24	120	322	461	407	182	34	2	0
	CDH74	16763	0	2	59	278	1145	3354	5373	4390	1811	337	14	0
	CDH80	7495	0	0	8	59	366	1424	2697	2142	708	90	1	0
Wind	WSAvg	7.7	7.9	8.4	9.3	9.8	8.2	7.7	6.4	5.9	6.4	7.3	7.8	7.6
Precipitation	PrecAvg	35.4	0.9	1.2	2.4	3.4	4.6	5.0	3.8	4.1	3.7	3.0	1.8	1.5
	PrecMax	60.9	2.7	3.5	8.4	8.7	11.8	15.2	11.0	11.2	12.7	7.2	6.3	4.3
	PrecMin	19.1	0.0	0.0	0.1	0.6	0.4	0.6	0.5	0.3	0.6	0.0	0.0	0.0
	PrecStd	8.1	0.6	0.9	1.7	1.9	2.3	2.9	2.8	2.6	2.6	1.9	1.6	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.7	72.1	80.1	86.1	91.5	97.4	102.5	102.9	96.5	87.5	76.2	66.5
		MCWB	51.2	54.4	62.4	64.8	71.5	76.1	75.6	75.6	73.9	69.9	60.6	57.4
	2%	DB	58.2	64.9	74.4	80.8	87.6	93.4	98.7	97.6	91.3	82.2	70.7	59.8
		MCWB	47.9	52.1	58.7	63.3	70.7	75.9	76.9	75.7	73.3	65.9	58.3	50.6
	5%	DB	52.6	59.1	69.6	77.0	84.3	90.7	95.5	94.3	87.7	77.4	65.8	54.7
		MCWB	44.3	49.2	56.9	61.6	69.1	75.0	77.1	75.5	72.3	64.3	54.4	47.3
	10%	DB	47.3	53.5	64.6	72.7	80.6	88.0	92.1	90.8	83.8	73.3	61.2	49.8
		MCWB	40.2	45.1	53.8	60.2	67.7	73.6	76.2	74.8	70.0	62.5	52.8	43.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.2	59.2	65.9	69.0	75.4	80.1	81.4	80.2	77.4	72.9	63.7	60.0
		MCDB	60.2	65.5	76.1	80.0	87.3	91.6	93.6	92.4	89.0	82.4	72.1	65.0
	2%	WB	48.4	54.0	62.3	66.7	73.3	78.1	79.7	78.6	75.3	69.8	60.1	53.3
		MCDB	56.4	61.7	70.2	76.6	83.6	89.7	92.2	91.4	86.5	77.7	67.5	57.5
	5%	WB	44.5	49.6	58.9	64.1	71.3	76.5	78.3	77.4	73.9	66.9	57.2	47.5
		MCDB	52.1	59.7	67.0	74.1	81.0	87.9	90.9	89.7	84.4	74.2	63.4	54.1
	10%	WB	40.5	44.7	54.6	61.2	69.2	75.0	77.3	76.0	72.2	63.8	53.4	43.5
		MCDB	47.0	53.1	64.1	71.0	78.3	85.6	89.6	87.2	81.8	71.0	60.3	50.0
Mean Daily Temperature Range	5% DB	MDBR	19.6	20.2	21.9	21.9	20.3	19.5	19.9	20.8	22.2	22.3	21.2	18.6
		MCDBR	29.4	30.4	30.1	28.1	24.1	21.6	22.9	24.7	24.3	26.9	28.4	27.2
		MCWBR	19.8	19.6	17.3	15.3	11.1	8.9	7.7	8.4	10.0	13.6	17.1	18.8
	5% WB	MCDBR	27.6	28.1	23.7	24.0	20.5	19.9	20.5	21.4	20.8	20.6	23.3	24.3
		MCWBR	19.6	19.4	15.2	15.0	11.2	9.5	8.5	8.9	9.8	12.5	16.4	18.7
Clear-Sky Solar Irradiance	taub		0.289	0.303	0.333	0.386	0.420	0.432	0.435	0.435	0.387	0.342	0.306	0.285
	taud		2.479	2.450	2.422	2.294	2.261	2.257	2.274	2.278	2.362	2.474	2.514	2.508
	Ebn at Noon		283	293	294	282	274	270	267	265	272	275	274	277
	Edh at Noon		25	29	34	41	43	43	42	41	35	29	24	23
All-Sky Solar Radiation	RadAvg		716	970	1255	1597	1830	2072	2056	1834	1523	1104	785	616
	RadStd		69	106	102	104	158	138	139	92	104	123	70	58

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+193	N/A
(47) Regional (26 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page